UNITED INTERNATIONAL UNIVERSITY

Department of Computer Science and Engineering (CSE)

CSE 3422: Software Engineering Laboratory



PROJECT REPORT

CareerGuide

A Full-Stack, Role-Based AI Career Guidance Platform

Prepared by

Tasbir Iqbal - 0112320001

Md. Shahriar Zaman Ayon - 0112330481

Atikur Rahman - 011221584

Taslima Tabassum Prova - 011222208

Md. Mostakim Foyzullah Khan - 0112310188

Afifa Tasnim - 011221087

Source Code Repository

github.com/Tasbir-Iqbal/Career_Guide_Project

2nd October 2026

Table of Contents

1. Introduction.....

2. Project Description.....

3. Screenshots of the Project.....

4. Technology Used.....

5. System Architecture and Workflows.....

6. Updated Project Structure.....

7. Local Setup and API Overview.....

8. Challenges and Future Work.....

9. Conclusion.....

1. Introduction

1.1 Project Overview

CareerGuide is a full-stack, role-based career guidance platform that helps students explore career paths, receive AI-powered recommendations, connect with approved counselors, book counseling sessions, join career webinars, and submit feedback. The platform also provides dedicated counselor and administrator portals for managing guidance activity, published content, and platform operations.

1.2 Objectives

- Provide students with personalized career suggestions based on education level, skills, interests, favorite subjects, and preferred work style.
- Use Google Gemini to generate structured career recommendations and detailed learning roadmaps.
- Provide secure role-based access for students, counselors, and administrators.
- Support human guidance through counselor discovery, session booking, and chat.
- Offer webinars, career articles, feedback moderation, and administrative reporting in one system.
- Preserve account and article history through soft deletion instead of permanent removal.

1.3 Motivation

Many students struggle to identify careers that match their strengths and interests, and individual guidance is not always accessible. CareerGuide AI combines structured self-assessment, AI assistance, and counselor support in one web application. The project also demonstrates practical software-engineering skills across frontend development, backend API design, relational databases, authentication, AI integration, and version control.

2. Project Description

The system serves three user roles. Each role receives only the functions needed for its responsibilities, and the backend enforces role-based authorization for protected endpoints. Only approved counselors can create webinars, only students can register for webinars, and only administrators can access reports and moderation functions.

Table 1: *Roles and access responsibilities*

Role	Main capabilities
Student	Career assessment, AI recommendations, detailed roadmap, counselor sessions, chat, webinars, and feedback.
Counselor	Session management, articles, chat, webinars, attendee viewing, and feedback.
Administrator	Users, counselor applications, categories, webinar oversight, feedback moderation, and reports.

2.1 List of Features

Student Portal

- Create an account and log in securely.
- Complete an AI Career Match assessment using education level, skills, interests, favorite subjects, and preferred work style.
- Receive three personalized career recommendations generated with Google Gemini.
- Select a preferred career path.
- View saved career paths and a personalized learning roadmap.
- Browse approved counselors and book counseling sessions.
- View and cancel eligible session bookings.
- Chat with counselors.
- Browse career webinars.
- Register for webinars and access meeting links only after registration.
- Submit feedback, suggestions, or complaints and track their review status.

Counselor Portal

- Register as a counselor and wait for administrator approval.
- Manage student counseling session requests.
- Confirm, complete, or cancel sessions.
- Publish and manage career-guidance articles.
- Chat with students.
- Create webinars linked to career categories.
- View webinar attendees and cancel future webinars.
- Submit feedback, suggestions, or complaints and track their status.

Administrator Portal

- View users, counselors, and system statistics.
- Create approved counselor accounts directly.
- Review and approve or reject counselor applications.
- Manage career categories.
- Oversee all webinars and cancel invalid or inappropriate sessions.
- Review feedback, suggestions, and complaints.
- Mark feedback as reviewed or resolved.
- Download a live CSV activity report containing user, session, webinar, registration, and feedback totals.

2.2 Description of Key Features

AI Career Match

A student enters education level, interests, skills, favorite subjects, and preferred work style. The FastAPI backend sends the profile to Google Gemini, which returns exactly three structured career recommendations. The backend validates the response and saves the assessment and recommendations in MySQL. The student can then choose one career path.

Detailed Career Roadmap

After selecting a career, the student can explicitly request a detailed AI roadmap. The roadmap includes the career goal, starting point, estimated duration, first seven days, staged learning actions, deliverables, readiness checks, portfolio projects, low-cost resources, common mistakes, and a next best action. The roadmap is stored in MySQL and reloaded later, avoiding unnecessary repeated AI requests.

Webinar Privacy

An approved counselor creates a webinar with a private meeting link. Students can browse webinar information but cannot see the link publicly. After a student registers, the meeting link becomes available in My Webinars. Administrators can oversee and cancel webinars but do not receive private meeting links.

Feedback Moderation

Students and counselors can submit feedback, suggestions, or complaints. Submissions begin with a new status, and administrators can filter them and mark them reviewed or resolved. The original sender can view the updated status in the portal.

Soft Deletion

Account and article deletion are implemented as soft deletion. The system stores is_deleted and deleted_at values rather than removing the row. Soft-deleted users cannot log in or continue using an old token. Soft-deleted articles are excluded from normal list and detail queries, and soft-deleted counselors are excluded from the public counselor list.

3. Screenshots of the Project

The following screens illustrate the main user journeys of the completed application.

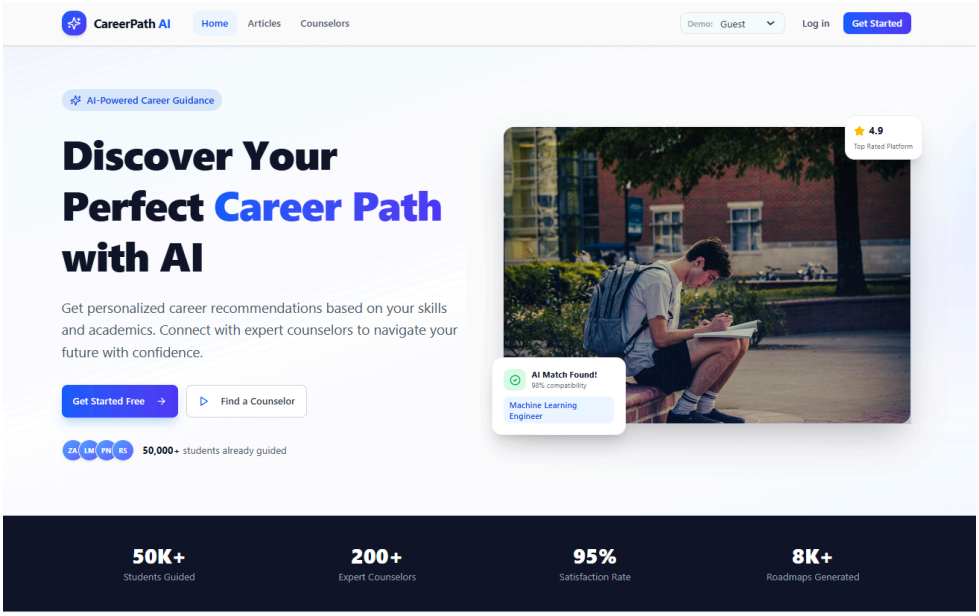


Figure 1: *Home Page*

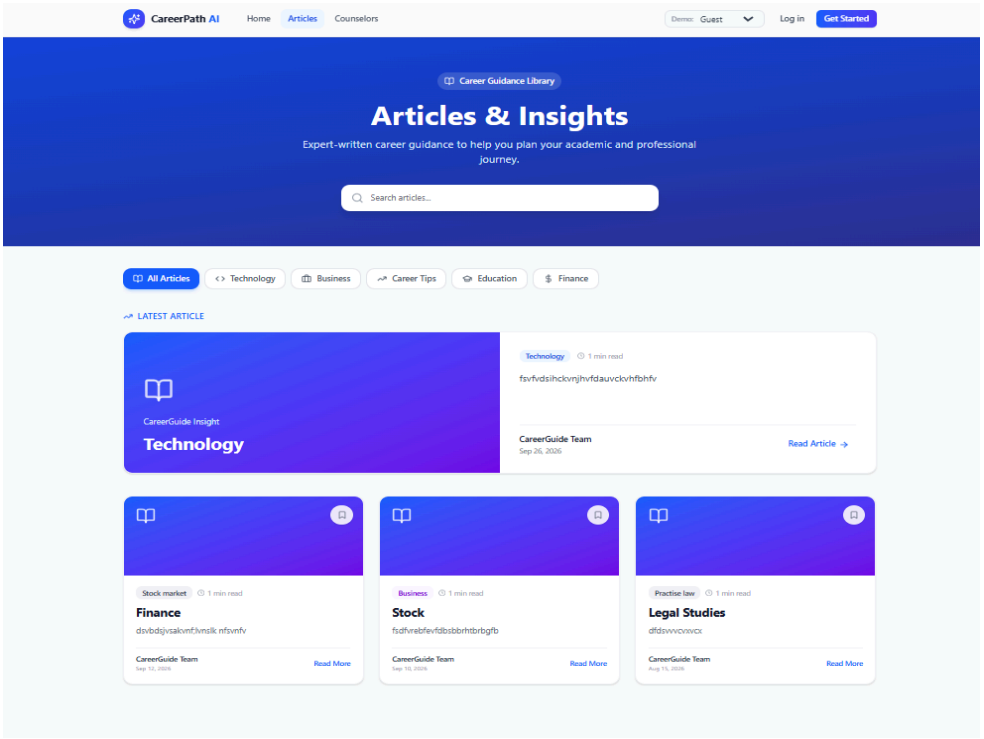


Figure 2: *Articles Page*

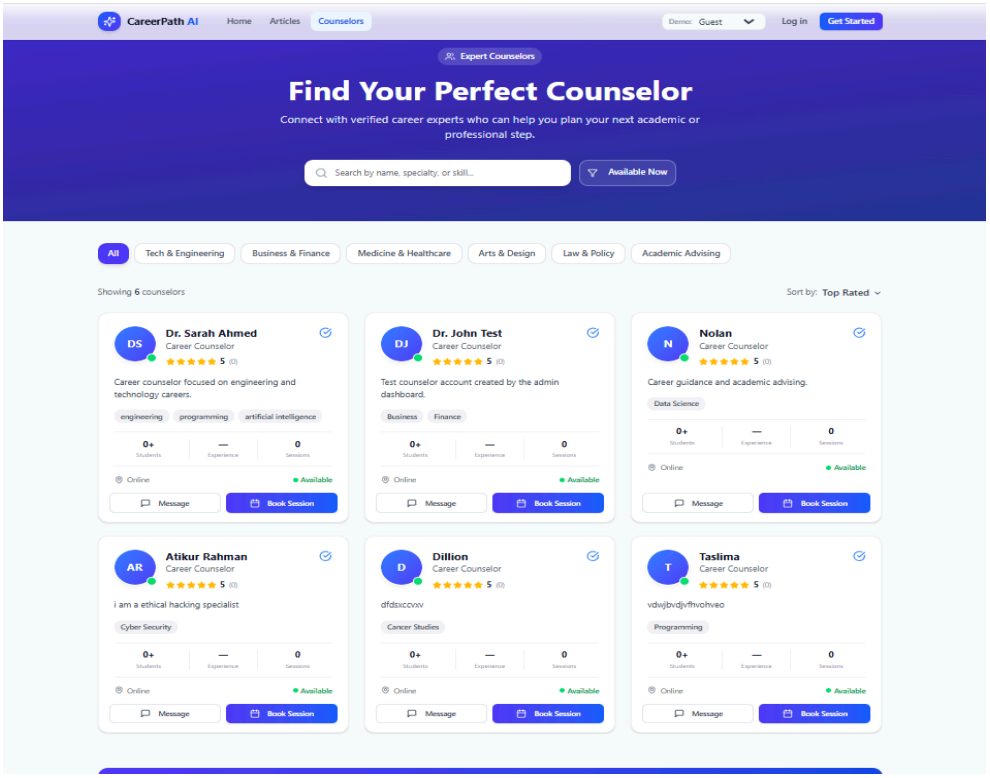


Figure 3: *Counselor Page*

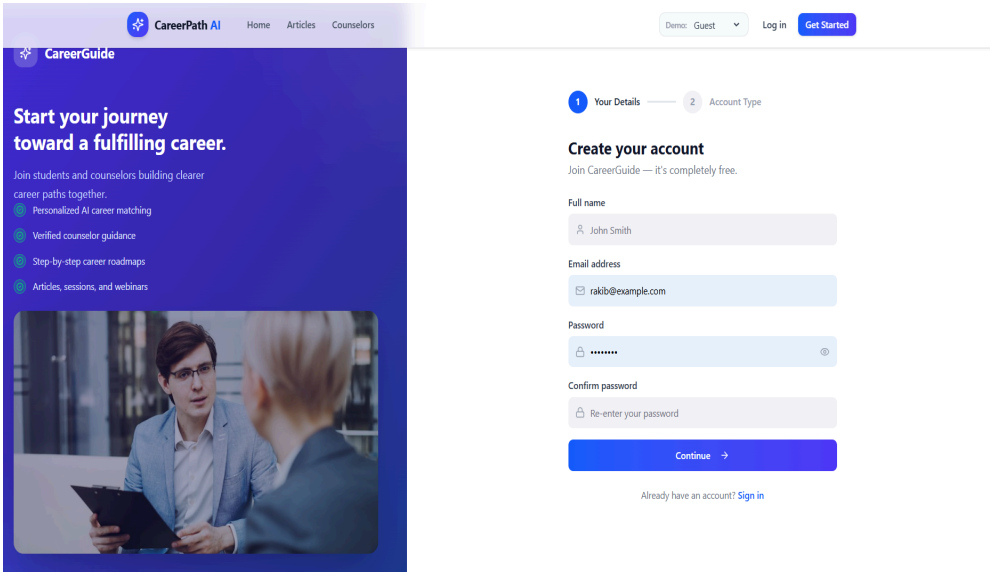


Figure 4: *Signup Page*

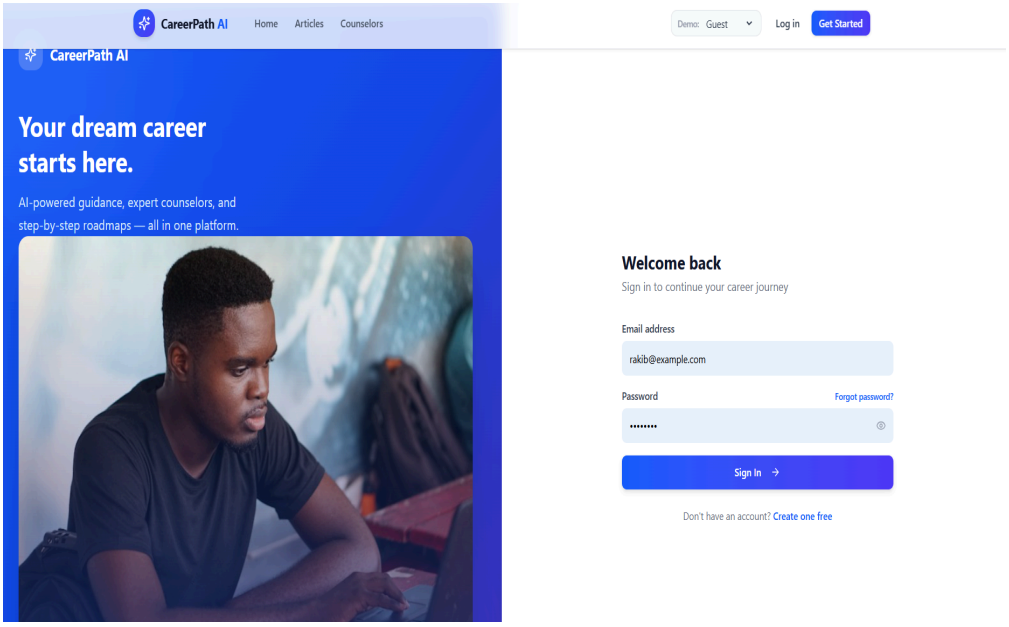


Figure 4: *Login Page*

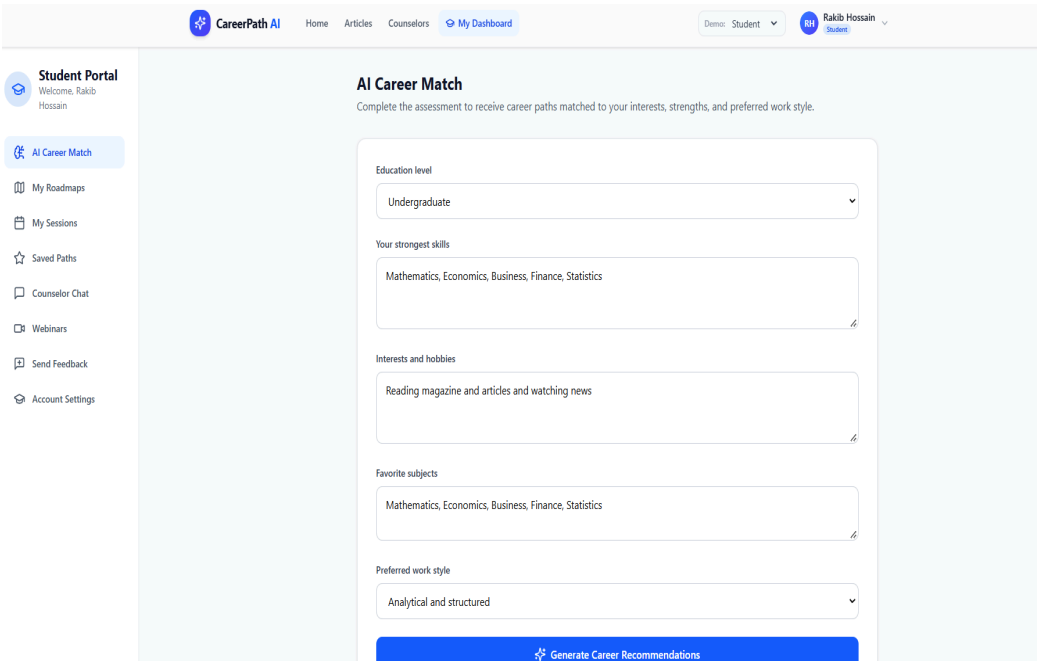


Figure 5: *Student Assessment*

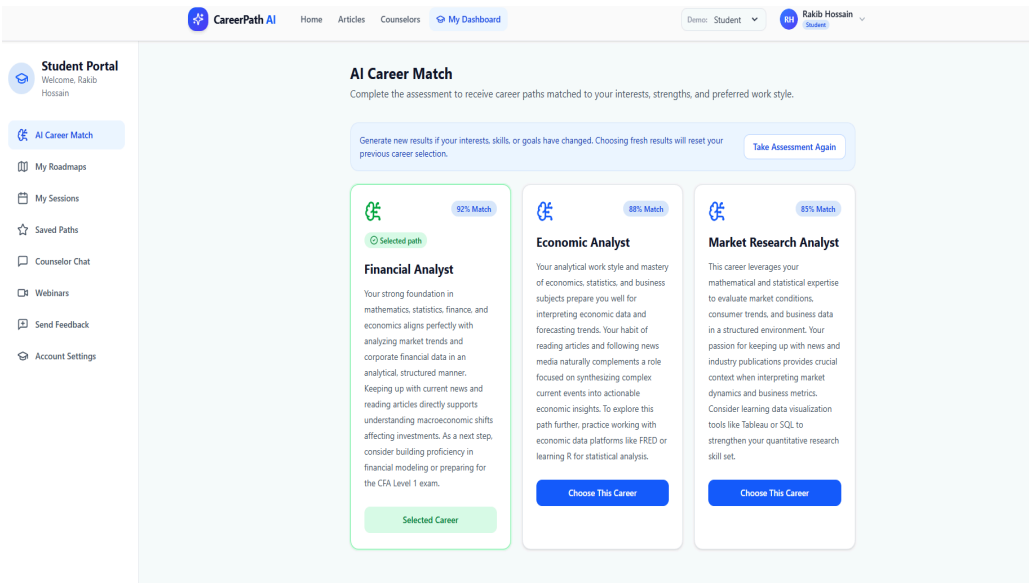


Figure 5: *AI Career Match*

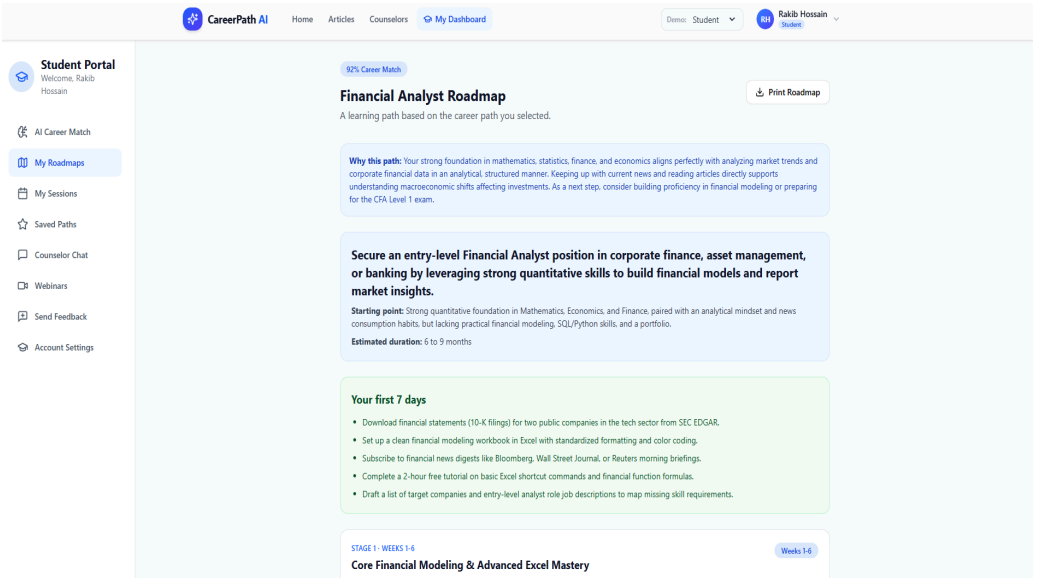


Figure 6: *Career Roadmap*

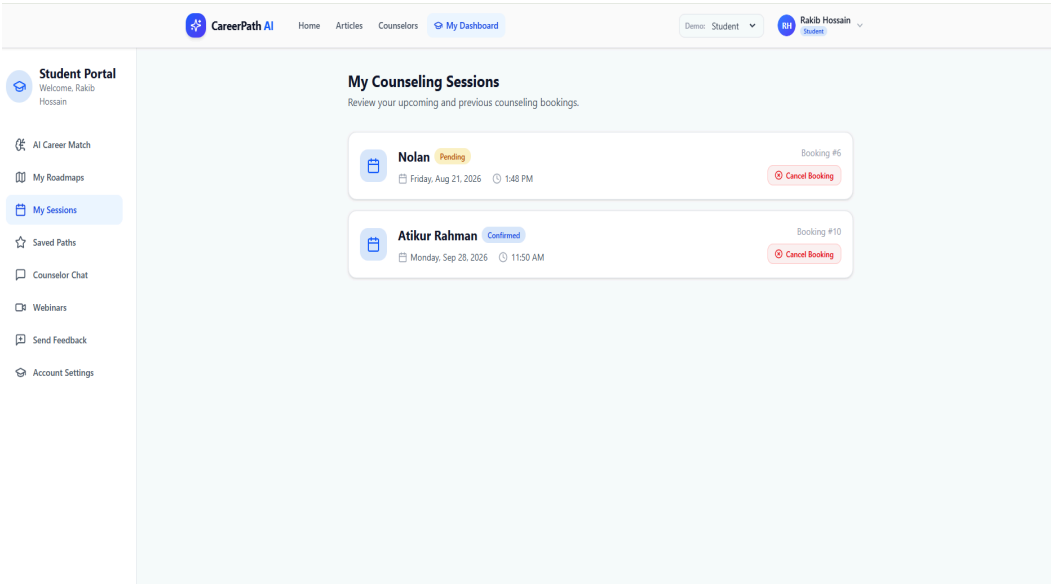


Figure 7: *Student Counseling Session*

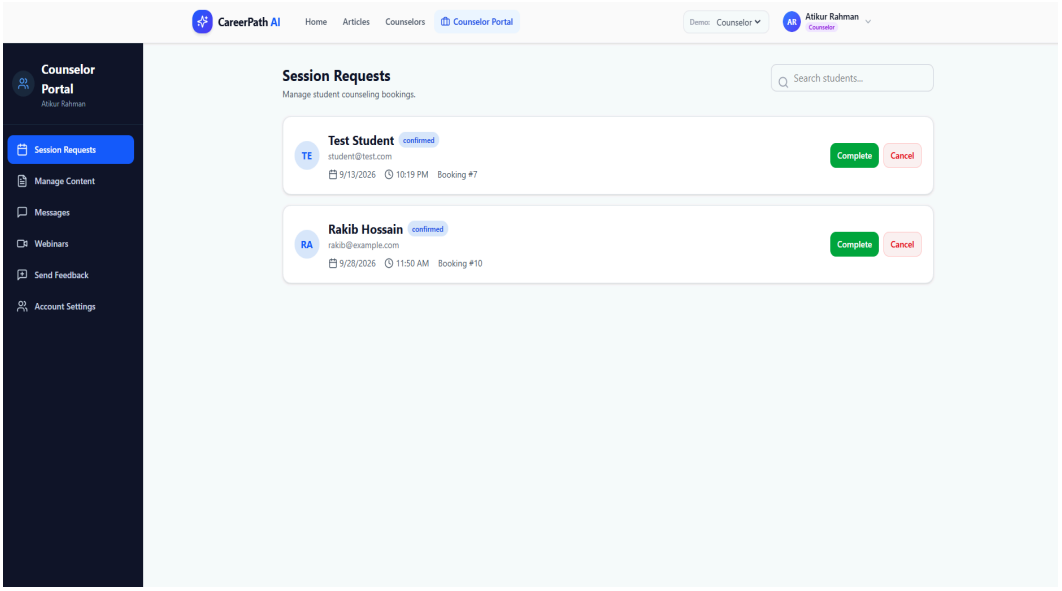


Figure 8: *Counselor Session Request*

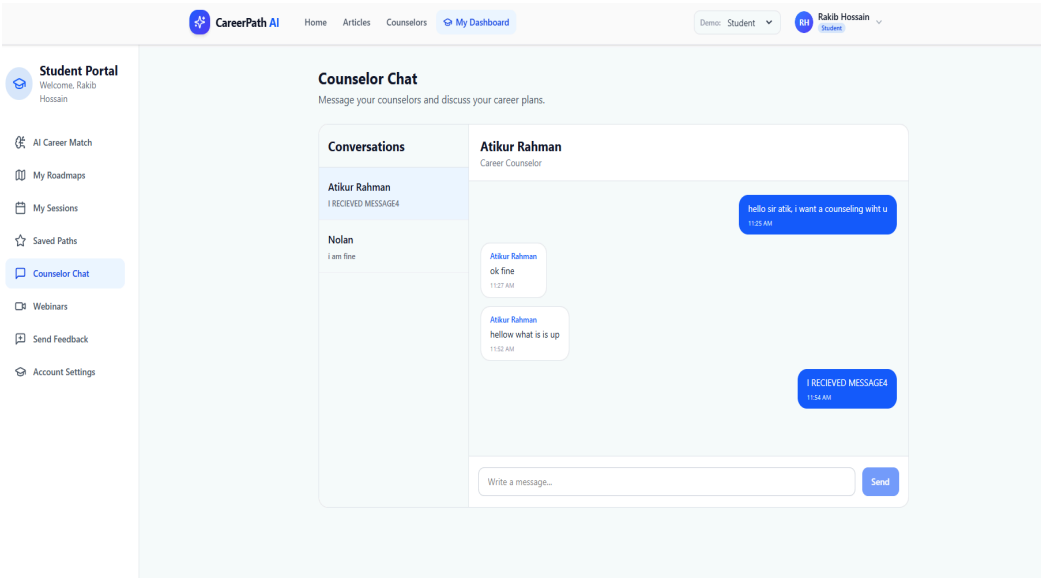


Figure 9: *Chat Interface*

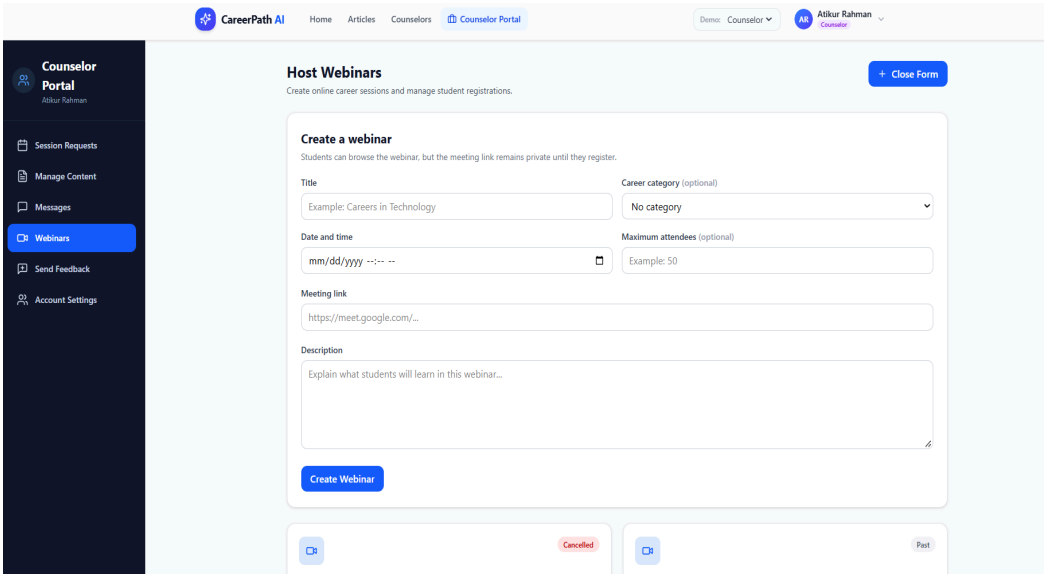


Figure 10: *Create Webinar*

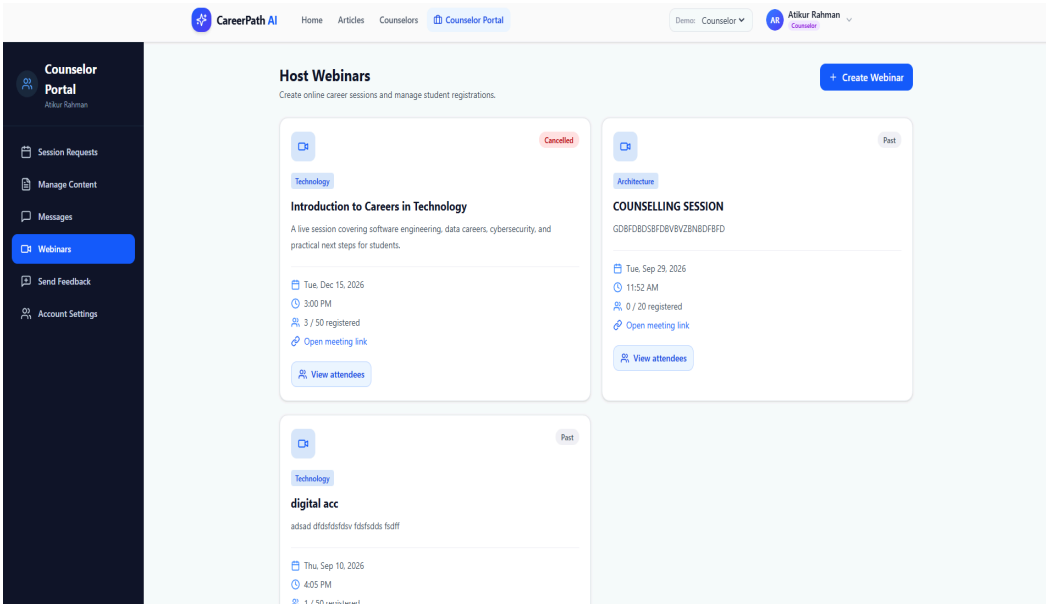


Figure 11: *Created Webinar Interface*

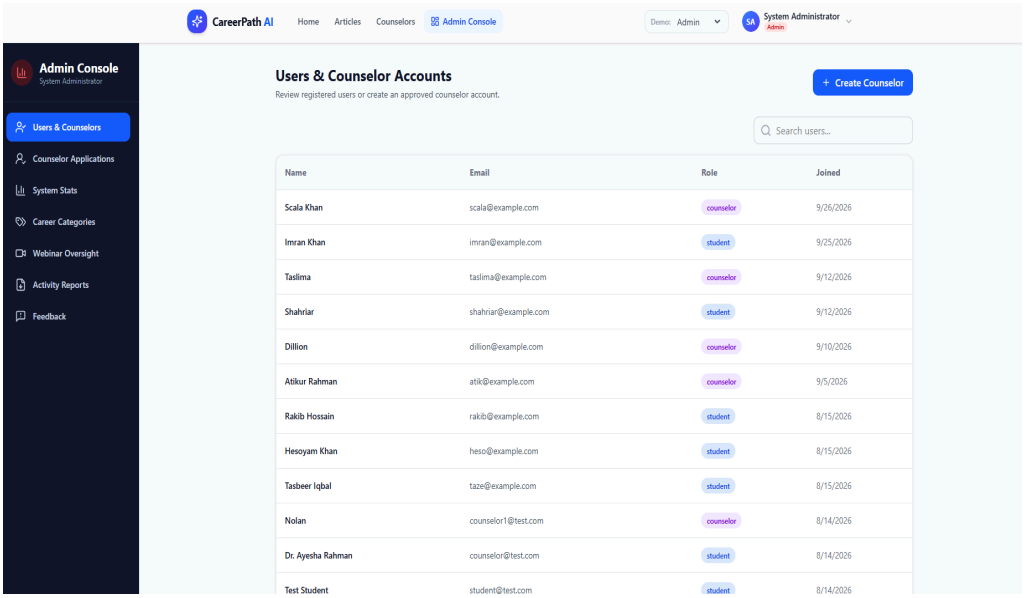


Figure 12: *Admin Console Review*

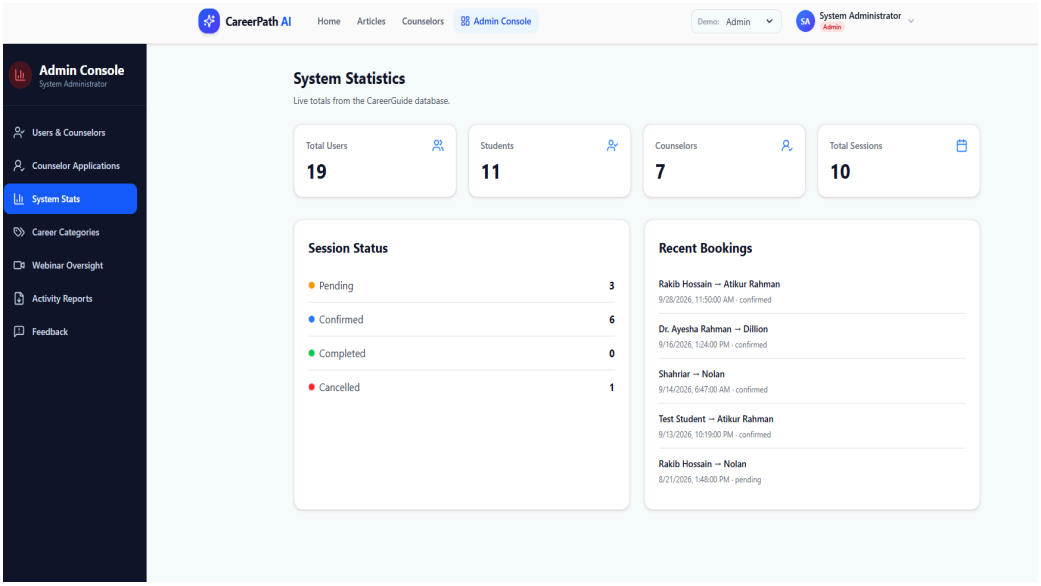


Figure 13: *Admin System Statistics*

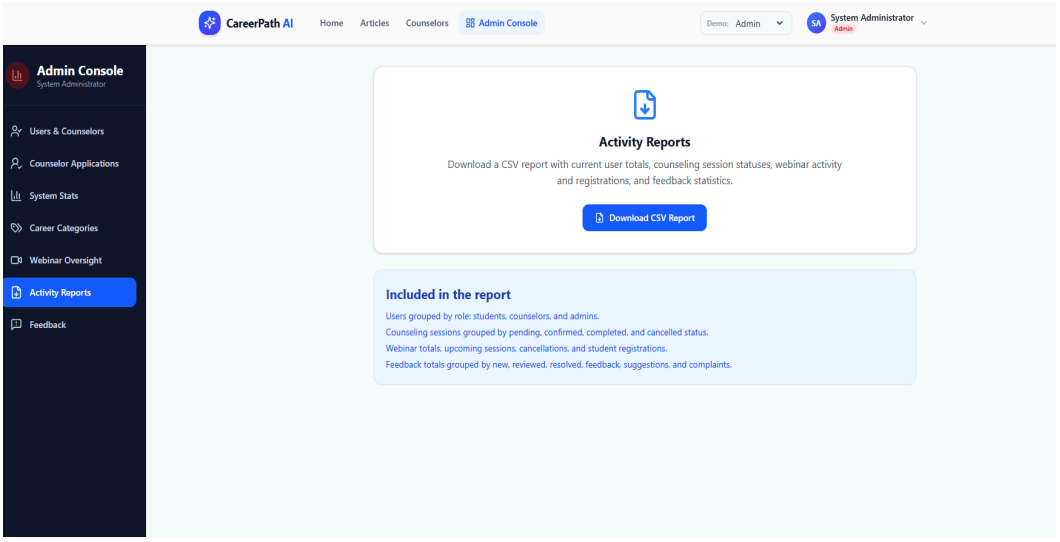


Figure 14: *Activity Report*

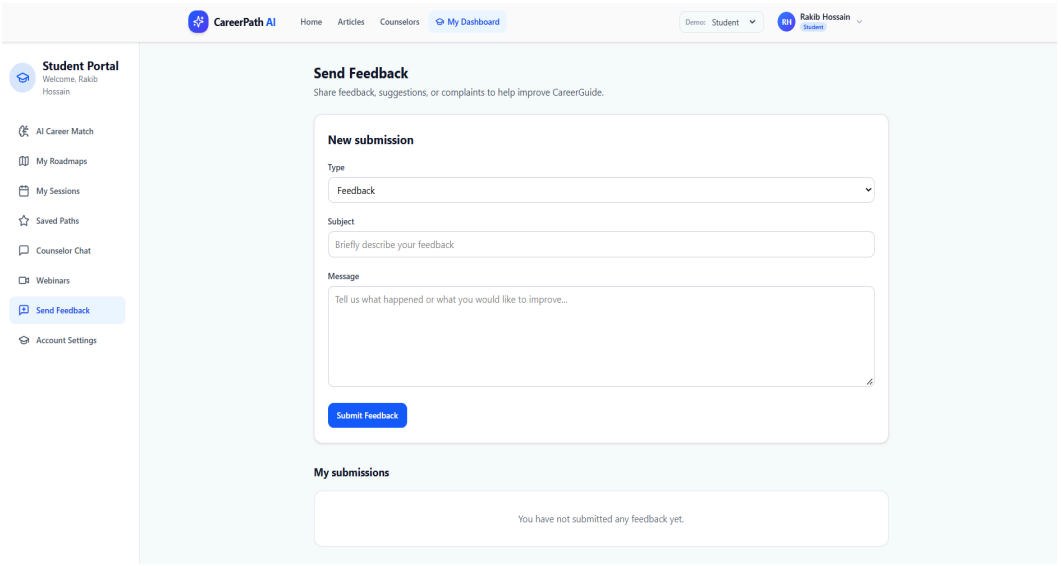


Figure 15: *Send Feedback*

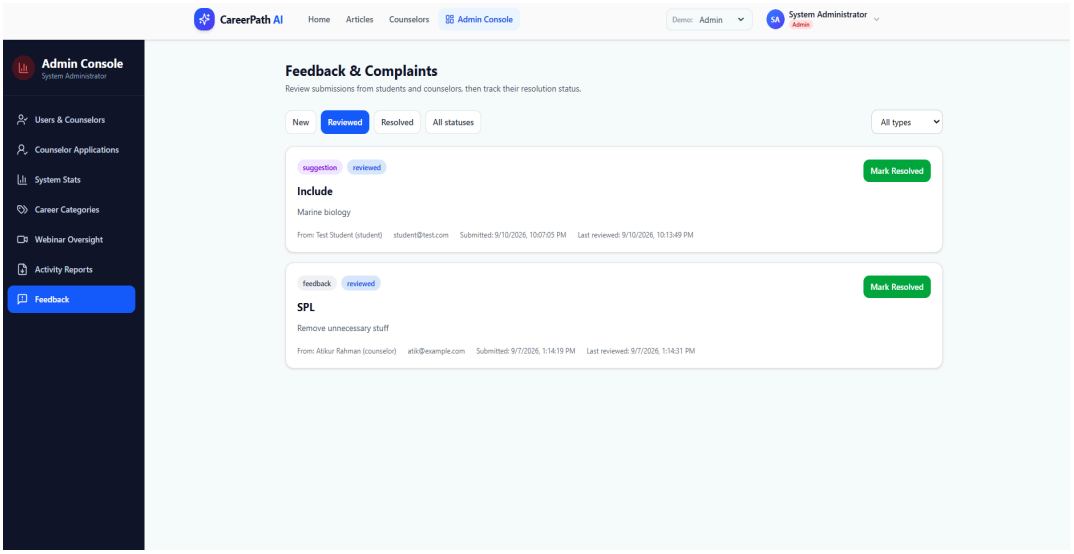


Figure 16: *Feedback Moderation Overview*

4. Technology Used

The project uses a modern web stack. React provides the user interface, FastAPI exposes the REST API, SQLAlchemy maps Python models to MySQL, and Google Gemini supplies AI-generated career guidance.

Table 2: *Technology stack*

Layer	Technology	Purpose
Frontend	React, TypeScript, Vite	Single-page application and role-based dashboards.
Styling	Tailwind CSS, Lucide React	Responsive layout, styling, and icons.
Backend	Python, FastAPI	REST endpoints, dependency injection, and API documentation.
Data access	SQLAlchemy, Pydantic	ORM models, relational queries, validation, and response schemas.
Database	MySQL	Persistent storage for users, sessions, recommendations, roadmaps, messages, webinars, feedback, and articles.
Authentication	JWT bearer tokens, password hashing	Secure login and role-based authorization.
AI model/API	Google Gemini API	Structured career recommendations and detailed roadmaps.
Reporting	FastAPI StreamingResponse, CSV	Live administrative activity report.
Version control	Git and GitHub	Source-code storage and project history.

4.1 Frontend

The frontend is a React and TypeScript application built with Vite. Separate dashboard pages serve students, counselors, and administrators. Components send authenticated API requests, manage loading and error states, and display role-appropriate data. Tailwind CSS is used for consistent styling.

4.2 Backend and APIs

FastAPI organizes the backend through routers for administration, articles, authentication, categories, chat, counselors, feedback, recommendations, students, and webinars. Pydantic schemas validate incoming and outgoing data. FastAPI also provides interactive Swagger documentation at /docs.

4.3 Database

MySQL stores the application data. Core entities include users, student profiles, counselor profiles, articles, career categories, session bookings, assessment attempts, recommendations, career roadmaps, conversations, messages, webinars, webinar registrations, and feedback.

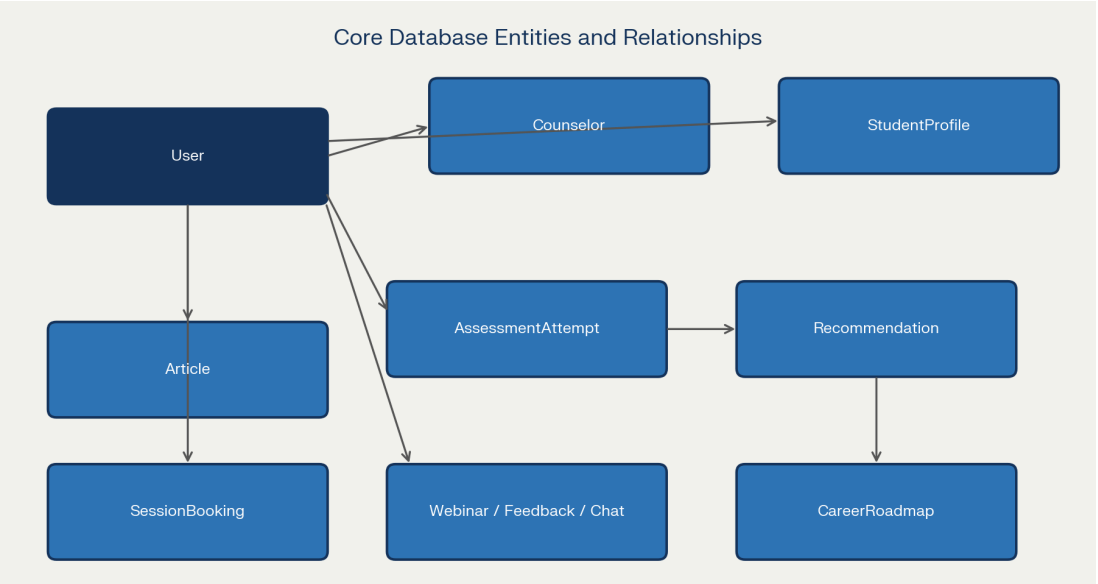


Figure 17: *Core Database Entities and Relationships (simplified)*

4.4 AI Model Integration

Google Gemini is called from the backend only. Structured JSON output allows the backend to validate recommendation and roadmap responses before saving them. Detailed roadmaps are generated when the student explicitly requests one, then stored and reused.

4.5 Authentication and Security

- Passwords are stored as hashes, not plaintext.
- JWT bearer tokens protect authenticated routes.
- Role-based authorization protects student, counselor, and administrator actions.
- Counselors require administrator approval before access.
- Meeting links are excluded from public webinar responses.
- Only registered students and the webinar owner can access meeting links.
- API keys and database credentials remain in backend environment variables.
- Soft-deleted users cannot log in or continue using older tokens.

4.6 Testing

- FastAPI Swagger documentation was used to inspect and test routes.
- The backend and dashboards were checked after database schema updates.
- Roadmap generation, persistence, refresh, and reload behavior were tested.
- Account soft deletion was verified by blocking later login and retaining the MySQL row.
- Article soft deletion was verified by hiding the article from the application while retaining the MySQL row.

4.7 Git and GitHub

Git and GitHub store the source code and project history. Sensitive files such as .env, virtual environments, node_modules, Python cache directories, API keys, and database credentials should be excluded from commits using .gitignore.

5. System Architecture and Workflows

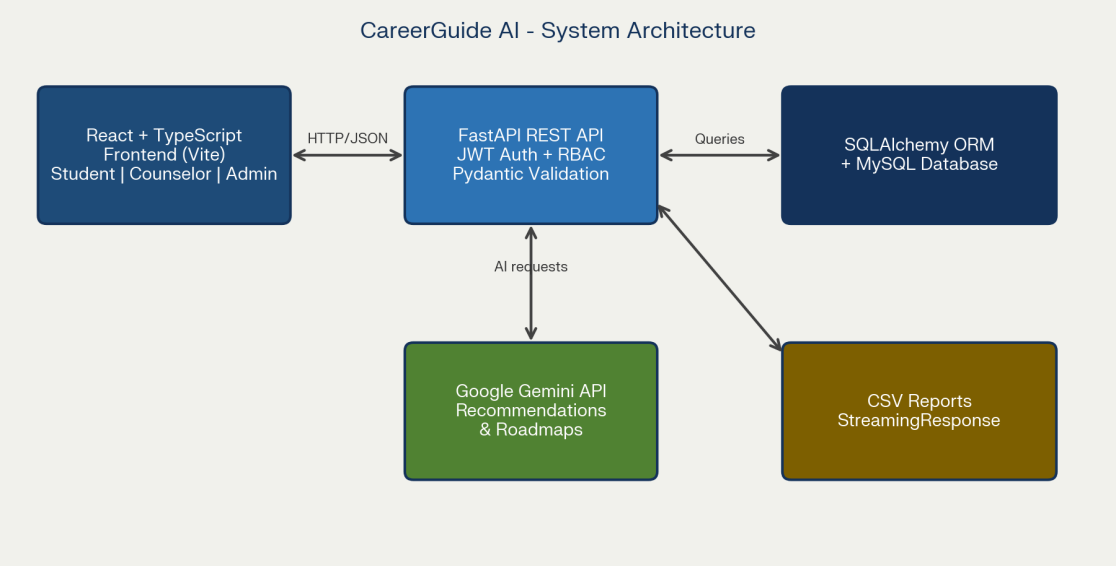


Figure 18: *System Architecture of CareerGuide AI*

The React frontend sends HTTP requests to FastAPI. The backend authenticates the user, enforces role-based permissions, reads or writes MySQL through SQLAlchemy, and calls Gemini only when AI-generated content is required.

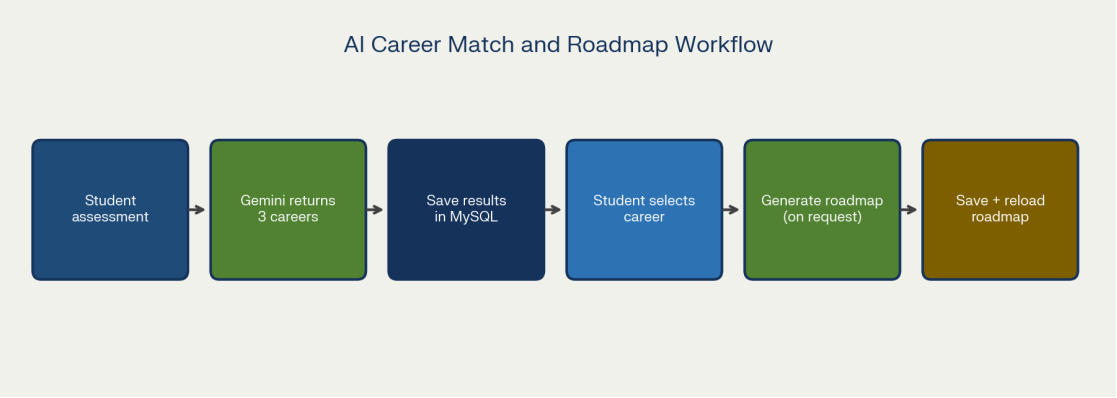


Figure 19: *AI Career Match and Detailed Roadmap Workflow*

5.1 AI Career Match Workflow

1. A student enters education level, interests, skills, favorite subjects, and work style.
2. The FastAPI backend sends the profile to Google Gemini.
3. Gemini returns exactly three structured career recommendations.
4. The backend validates the results and saves the assessment and recommendations in MySQL.
5. The student chooses one career path for saved paths and roadmap guidance.
6. The student requests a detailed roadmap, which is generated once and saved for later retrieval.

5.2 Webinar Privacy Workflow

1. An approved counselor creates a webinar with a private meeting link.
2. Students browse webinar information but cannot see the meeting link.
3. A student registers for the webinar.
4. The registered student accesses the link in My Webinars.
5. Administrators oversee and cancel webinars but do not receive private meeting links.

5.3 Feedback Moderation Workflow

1. Students and counselors submit feedback, suggestions, or complaints.
2. Submissions are stored with a new status.
3. Administrators filter submissions and mark them reviewed or resolved.
4. The original sender can view the updated status.

6. Project Structure

The structure below reflects the current GitHub repository. The frontend source and Vite configuration are located directly in Career_Guide_Project_100%, while the backend is located in the BackEnd subfolder.

```
Career_Guide_Project/
├── README.md
│
└── Career_Guide_Project_100%/
    ├── ATTRIBUTIONS.md
    ├── README.md
    ├── index.html
    ├── package.json
    ├── package-lock.json
    ├── postcss.config.mjs
    ├── vite.config.ts
    │
    ├── src/
    │   ├── components/
    │   ├── context/
    │   │   └── AuthContext.tsx
    │   ├── pages/
    │   │   ├── AdminDashboard.tsx
    │   │   ├── CounselorDashboard.tsx
    │   │   ├── StudentDashboard.tsx
    │   │   └── ...
    │   ├── App.tsx
    │   ├── main.tsx
    │   └── ...
    │
    ├── guidelines/
    │   └── ...
    │
    └── BackEnd/
        ├── app/
        │   ├── routers/
        │   │   ├── __init__.py
        │   │   ├── admin.py
        │   │   ├── articles.py
        │   │   ├── auth.py
        │   │   ├── categories.py
        │   │   ├── chat.py
        │   │   ├── counselors.py
        │   │   ├── feedback.py
        │   │   ├── recommendations.py
        │   │   ├── students.py
        │   │   └── webinars.py
        │   ├── __init__.py
        │   ├── auth.py
        │   ├── database.py
        │   ├── gemini_service.py
        │   ├── main.py
        │   ├── models.py
        │   └── schemas.py
        ├── create_test_admin.py
        ├── create_test_counselor.py
        └── test_models.py
```

Figure 20: *Repository Folder Structure*

7. Local Setup and API Overview

7.1 Prerequisites

- Python 3.11 or newer
- Node.js 18 or newer
- MySQL Server
- A Google Gemini API key

7.2 Clone the Repository

```
git clone https://github.com/Tasbir-Iqbal/Career_Guide_Project.git
cd Career_Guide_Project/Career_Guide_Project_100%
```

7.3 Start the Backend

```
cd BackEnd
python -m venv venv
.\venv\Scripts\Activate.ps1
pip install -r requirements.txt
.\venv\Scripts\python.exe -m uvicorn app.main:app --reload
```

Create the MySQL database and a local .env file containing database credentials, JWT secret, Gemini API key, and Gemini model name. Never commit the real .env file. The backend runs at http://127.0.0.1:8000, and Swagger documentation is available at http://127.0.0.1:8000/docs.

7.4 Start the Frontend

```
cd Career_Guide_Project/Career_Guide_Project_100%
npm install
npm run dev
```

The frontend commonly runs at http://localhost:5173.
If required, set VITE_API_URL=http://127.0.0.1:8000 in the frontend environment file

7.5 API Highlights

Table 3: *Selected API endpoints*

Area	Example endpoints
Authentication	POST /auth/signup, POST /auth/login, GET /auth/me, DELETE /auth/me
Recommendations	POST /recommendations/generate, GET /recommendations/me
Detailed roadmap	POST /recommendations/{id}/roadmap, GET /recommendations/{id}/roadmap
Sessions	POST /students/sessions, GET /counselors/me/sessions
Chat	GET /chat/conversations, POST /chat/conversations/{id}/messages
Categories	GET /categories, POST /categories
Webinars	GET /webinars, POST /webinars, POST /webinars/{id}/register
Feedback	POST /feedback, GET /feedback/mine, GET /feedback/admin/all
Reports	GET /admin/reports/activity.csv

8. Challenges and Future Work

8.1 Problems Faced and Solutions

Table 4: *Challenges and solutions*

Problem	Solution
Different roles needed different permissions.	Role-based authorization and protected FastAPI dependencies.
AI output had to be reliable for the interface.	Structured JSON output, backend validation, and saved results.
Repeated AI requests could waste quota and time.	Generate the detailed roadmap on request, store it, and reload it from MySQL.
Webinar links needed protection.	Hide links from public responses and reveal them only to permitted users.
Permanent deletion could remove historical records.	Implement soft deletion with is_deleted/deleted_at and filter deleted data from active queries.
Documentation drifted from the real repository layout.	Update the README and report to match the current GitHub structure.

8.2 Upgrades and New Features

- Add Alembic database migrations instead of relying only on create_all.
- Add rate limiting, especially for login and Gemini endpoints.
- Add email notifications for counselor approval, bookings, webinar registration, and feedback updates.
- Add pagination, search, and date filters for large administrative datasets.
- Add automated tests for API authorization, webinar privacy, soft deletion, and role boundaries.
- Add an administrator interface for restoring soft-deleted accounts and articles.
- Add production deployment configuration with HTTPS and restricted CORS origins.
- Add PDF report export alongside CSV download.
- Add roadmap progress tracking and analytics.

9. Conclusion

CareerGuide is a complete software-engineering project that integrates a React frontend, FastAPI backend, MySQL database, JWT authentication, role-based authorization, Google Gemini, and administrative reporting. It guides students from career assessment to saved roadmap generation while also supporting counselors, webinars, chat, articles, feedback, and administrator oversight. Soft deletion strengthens data safety by retaining records while hiding deleted accounts and articles from normal use. The platform provides a strong foundation for further testing, analytics, notifications, migrations, and production deployment.